

CPU/Cache Optimization in Elliptic Curve Search: Performance Analysis

Theoretical Execution Model

Let N_h be the thread count, N_a the number of precomputed combinations (anomalies × channels), and $M = i_{\text{end}} - i_{\text{start}} + 1$ the range of linear iterations.

Total execution time decomposes as:

$$T_{\text{total}} = \frac{1}{N_h} \sum_{w=1}^{N_h} \left(T_{\text{comp}}^{(w)} + T_{\text{mem}}^{(w)} + T_{\text{sync}}^{(w)} \right)$$

where:

- $T_{\text{comp}}^{(w)} = N_a \cdot M \cdot C_{\text{key}}$ (cycles for key generation + hashing)
- $T_{\text{mem}}^{(w)} = N_a \cdot M \cdot (p_{\text{miss}} \cdot L_{\text{miss}} + (1 - p_{\text{miss}}) \cdot L_{\text{hit}})$
- $T_{\text{sync}}^{(w)} \approx N_a \cdot C_{\text{atomic}}$ (per-thread **found** flag checks)

Cycles per key check (C_{key}):

$$C_{\text{key}} = C_{\text{combine}} + C_{\text{serialize}} + C_{\text{SHA256}} + C_{\text{RIPEMD160}} + C_{\text{memcmp}} \approx 1200 + 300 + 800 + 600 + 50 = 2950 \text{ cycles}$$

Cache miss penalty:

$$L_{\text{miss}} \approx 200 \text{ cycles}, \quad L_{\text{hit}} = 4 \text{ cycles}$$

Cache miss probability after optimisation:

$$p_{\text{miss}} \approx \frac{D_{\text{working set}}}{C_{\text{cache}}} \cdot \alpha, \quad \alpha \in [0.05, 0.15] \text{ (due to prefetching and locality)}$$

With $D_{\text{working set}} = N_a \cdot |\text{Precomp}|$ and a 16 MB L3 cache, p_{miss} drops from ~0.6 to ~0.07.

Memory Structures and Alignment

The precomputation table is defined as:

```
Struct Precomp {
    secp256k1_pubkey const_pub; // 64 bytes (compressed)
    uint64_t anom_offset;      // 8 bytes
    int ch;                    // 4 bytes
    const char* name;          // 8 bytes
    uint128_t const_value;     // 16 bytes
    char padding[28];          // pad to 64 bytes total
} __attribute__((aligned(64)));
```